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1. Introduction

In today's educational environment, computer networks play a critical role in enabling efficient communication and information exchange within universities and other academic institutions. A well-designed network infrastructure allows multiple departments and campuses to share data seamlessly while also improving operational efficiency, security, and centralized resource management across the institution. This project focuses on the design and implementation of a multi-campus university network using Cisco Packet Tracer. The proposed network includes four campuses located in Karachi, Lahore, Quetta, and Peshawar, all interconnected through routers using the Routing Information Protocol (RIP) dynamic routing protocol. Each campus consists of five departments, and internal communication within campuses is managed through switches to ensure structured and efficient data flow.

LAN (Local Area Network) is a network that connects devices within a small area such as a room, office, building, or university department. It allows devices like PCs, laptops, printers, and CCTV cameras to communicate and share data with each other.

WAN (Wide Area Network) is a network that connects multiple LANs over large distances, such as between different university campuses or cities. It helps in communication and data transfer between different locations.

In this project, multiple devices such as PCs, laptops, printers, CCTV cameras, and wireless access points are connected to create a realistic university network environment. Static IP addressing and subnetting are used to organize and manage the network efficiently. A /28 subnet mask is applied for LAN segmentation, while /30 subnetting is used for WAN links between routers to utilize IP addresses efficiently. The main objective of this project is to design a secure, scalable, and efficient university network that supports smooth communication between departments and campuses. It also provides practical understanding of important networking concepts such as IP addressing, subnetting, routing protocols, and wireless communication, helping to connect theoretical knowledge with real-world implementation.

1.1 Overview of University Network

The university network designed in this project consists of four campuses:

- Karachi Campus
- Lahore Campus
- Quetta Campus
- Peshawar Campus

Each campus contains five departments:

- Administration Department
- Library Department
- Examination Department
- Accounts Department
- Education Department

Each department is connected through a dedicated switch, ensuring proper segmentation and efficient management of the local network. A range of end devices, including PCs, laptops,

printers, and CCTV cameras, are connected to these switches to support day-to-day departmental operations. In selected campuses such as Lahore and Quetta, wireless access points are also deployed to provide Wireless Fidelity (WiFi) connectivity for wireless-enabled devices, allowing both wired and wireless access within the same infrastructure.

All campuses are interconnected using Router-2811 through serial WAN links, forming the backbone of inter-campus communication. The RIP dynamic routing protocol is implemented to enable automatic route exchange between all campuses, ensuring that data can be transmitted across the network without manual routing configuration. Overall, the system is designed to support reliable communication between all devices across multiple campuses, representing a structured and real-world university networking environment.

1.2 Objectives of the Project

The main objectives of this project are:

- To design a multi-campus university network using Cisco Packet Tracer
- To understand the practical implementation of computer networking concepts
- To configure routers, switches, and wireless access points
- To implement static IP addressing for all devices
- To apply subnetting for better IP address management
- To establish communication between campuses using RIP routing protocol
- To provide wired and wireless communication in the network
- To test connectivity between departments and campuses
- To improve understanding of network design and configuration

1.3 Scope of the Project

The scope of this project is limited to the design and simulation of a multi-campus university network using Cisco Packet Tracer. The network includes both LAN and WAN connectivity across multiple campuses and departments, enabling structured communication within and between different locations. It supports inter-campus communication through dynamic routing to ensure efficient and automated data exchange. Wireless connectivity is also implemented in selected campuses using access points secured with WiFi authentication. Various networking devices such as routers, switches, PCs, laptops, printers, and CCTV cameras are configured to simulate a realistic university environment.

Future improvements to this project may include the integration of advanced networking technologies to enhance performance and scalability. These enhancements can involve VLAN segmentation, dedicated servers, Dynamic Host Configuration Protocol (DHCP) services, improved network security mechanisms, and cloud-based infrastructure to make the system more efficient and closer to real-world enterprise-level university networks.

2 Background Concepts

This project is based on different networking concepts that are commonly used in real-world computer networks. These concepts help in understanding how devices communicate with each other, how data travels through the network, and how different campuses stay connected. The

following concepts were used during the design and implementation of this university network project.

2.1 Computer Networking Basics

Computer networking is the process of connecting multiple devices so they can communicate and share data efficiently. It enables devices such as computers, laptops, printers, and servers to exchange information through both wired and wireless communication channels, forming the foundation of modern digital communication systems. In universities, offices, banks, hospitals, and other organizations, computer networking plays a critical role in improving operational efficiency by allowing users to share resources, communicate in real time, and access centralized information systems with greater speed and reliability, making organizational processes more coordinated and effective.

Different networking devices are used to build a network, such as:

- Routers
- Switches
- Access Points
- PCs and Laptops
- Printers
- CCTV Cameras

In this project, routers are used to connect different campuses, switches are used to connect departmental devices, and access points are used to provide wireless communication.

2.2 Internet Protocol (IP) Addressing

An IP address is a unique identifier assigned to each device within a network, enabling proper identification and communication between connected devices. Without IP addressing, devices would not be able to exchange data accurately, making network communication impossible.

An IPv4 address is represented as four numerical octets separated by dots, such as 192.168.10.2. In this project, static IP addressing is used, where each device is manually configured with a unique IP address. This includes end devices such as PCs, laptops, printers, CCTV cameras, and network devices like routers to ensure proper and conflict-free communication across the network.

IP addressing helps in:

- Device identification
- Communication between networks
- Data transmission
- Network management

The IP address 192.168.10.2 can also be explained as follows:

- **192** → Represents the network class or major network portion. In private networks, 192 is commonly used for local area networks (LANs).

- **168** → Further identifies the specific private network. Together, 192.168 shows that the address belongs to a private IP range used inside organizations, homes, and campuses.
- **10** → Represents the subnet or department network. It helps divide the main network into smaller logical sections for better management and security.
- **2** → Represents the host or device number. It identifies the specific device within that subnet. For example, this may be assigned to a PC, printer, router interface, or CCTV camera.

In this example:

- **192.168.10** → Network/Subnet portion
- **2** → Host/Device portion

This structure allows devices to communicate properly while helping network administrators manage the network efficiently.

2.3 Class C IP Addressing

Class C IP addresses are commonly used in small and medium-sized networks. In this project, private Class C IP addresses are used for all campus networks.

The Class C IP range starts from: **192.0.0.0 to 223.255.255.255**

The network uses private IP ranges such as:

- **192.0.10.0**
- **192.0.20.0**
- **192.0.30.0**
- **192.0.40.0**

These IP addresses are divided into smaller subnets for different departments. Using Class C addressing makes the network easier to manage and organize.

2.4 Subnetting Concept

Subnetting is the process of dividing a large network into smaller, more manageable networks known as subnets, which improves network efficiency by enhancing performance, security, and overall administration of network resources. In this project, subnetting is implemented to create separate logical networks for each department, where each department is assigned its own subnet along with a dedicated default gateway to ensure structured communication and reduce unnecessary network traffic between different segments.

For departmental Local Area Networks (LANs) the subnet mask is used: /28=255.255.255.240

A /28 subnet mask (255.255.255.240) is used for LAN networks inside each campus. In a /28 subnet, the total number of IP addresses is 16, out of which 14 are usable for devices. This

subnet is suitable for departments because each department contains a limited number of devices such as PCs, laptops, printers, and CCTV cameras. It provides enough IP addresses without wasting resources and keeps the network properly segmented.

The /28 subnet can be explained as follows:

- **255.255.255.240**
- **255.255.255** → Represents the network portion. These parts remain fixed for all devices within the same network.
- **240** → Represents the subnet portion.
- In binary form, 240 is written as: **11110000**
- This means the first 4 bits are used for the network, while the remaining 4 bits are available for host devices.
- Equation for Total Hosts
 - Formula for Calculating Total Hosts: 2^h
 - Here "h" represents host bits. If it's a /28 subnet, then 4 bits are available for hosts.
 - So the Total IP addresses is $2^4 = 16$
- Equation for Usable Hosts:
 - Formula for Usable Hosts: $2^h - 2$
 - The formula for usable hosts is $2^h - 2$ because there are 2^h total IP addresses and out of them 2 are reserved (network and broadcast), hence they are subtracted.
 - So the Usable host addresses is $2^4 - 2 = 14$

Example:

- Network Address → 192.168.10.0
- First Usable IP → 192.168.10.1
- Last Usable IP → 192.168.10.14
- Broadcast Address → 192.168.10.15

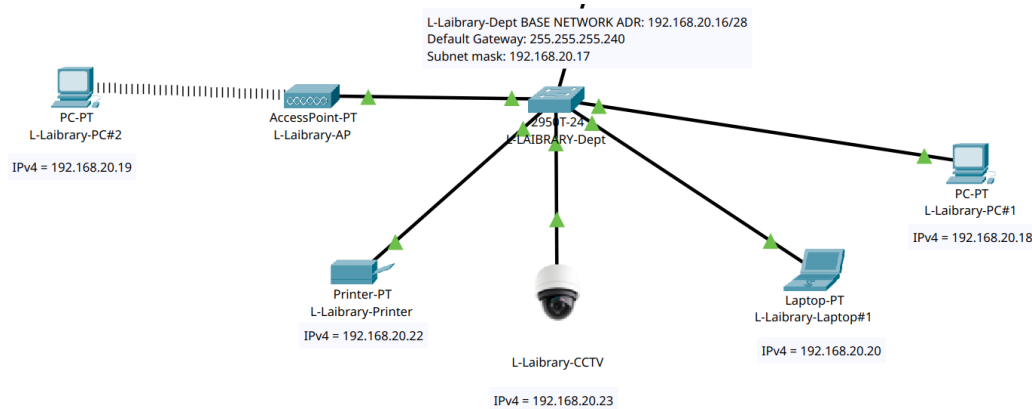


Figure 01: LAN Subnetting /28

For router-to-router WAN communication the subnet mask is used: /30 = 255.255.255.252

A /30 subnet mask (255.255.255.252) is used for WAN connections between routers. In a /30 subnet, the total number of IP addresses is 4, out of which only 2 are usable. This subnet is used for point-to-point links between routers, where only two devices communicate directly. It is highly efficient because it avoids IP address wastage and is ideal for inter-campus connections.

The /30 subnet can be explained as follows:

- **255.255.255.252**
- **252** in binary form is: **11111100**
- This means 6 bits are reserved for the network and only 2 bits remain for host devices.
- Equation for Total Hosts
 - Formula for Calculating Total Hosts: 2^h
 - Here "h" represents host bits. If it's a /30 subnet, then 2 bits are available for hosts.
 - So the Total IP addresses is $2^2 = 4$
- Equation Usable Host
 - Formula for Usable Hosts: $2^h - 2$
 - The formula for usable hosts is $2^h - 2$ because there are 2^h total IP addresses and out of them 2 are reserved (network and broadcast), hence they are subtracted.
 - So the Usable host addresses is $2^2 - 2 = 2$

Example:

- Network Address → 192.168.50.0

- Usable IPs → 192.168.50.1 and 192.168.50.2
- Broadcast Address → 192.168.50.3

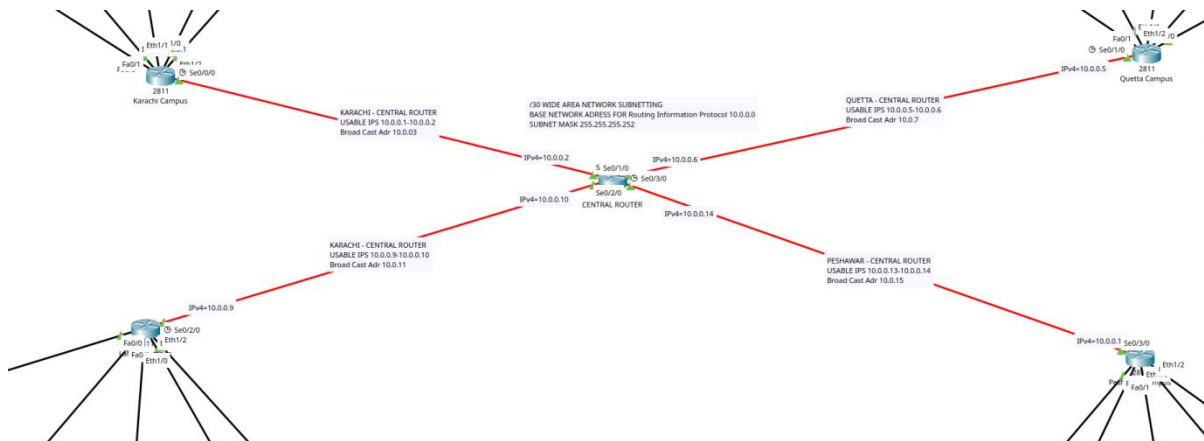


Figure 02: WAN Subnetting /30

2.5 Static IP Addressing

Static IP addressing refers to the manual assignment of fixed IP addresses to network devices, where each device retains the same IP address unless it is changed by the network administrator. This approach ensures stable communication, predictable connectivity, and easier management of important network devices. In this project, static IP addresses are assigned to routers, printers, CCTV cameras, and several end devices to maintain reliable communication across the university network infrastructure.

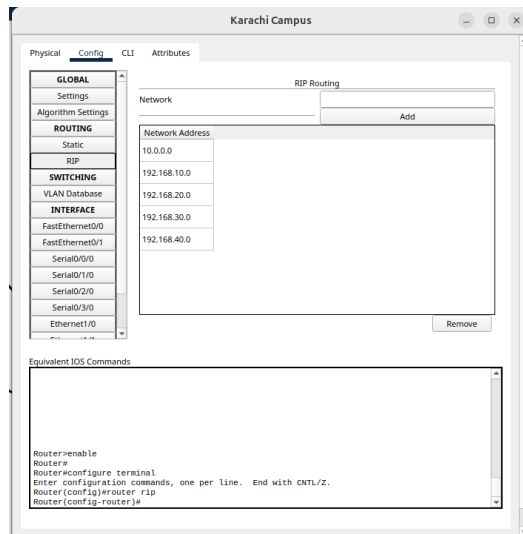
Along with static addressing, Dynamic Host Configuration Protocol (DHCP) is also implemented in selected departments to provide automatic IP address assignment for end-user devices. Through DHCP, connected devices automatically receive IP addresses, subnet masks, and default gateway information from the configured DHCP service. The combination of static and dynamic IP addressing improves network flexibility, reduces manual configuration effort, minimizes IP conflicts, and supports efficient communication within the university network.

2.6 Dynamic Routing using Routing Information Protocol (RIP)

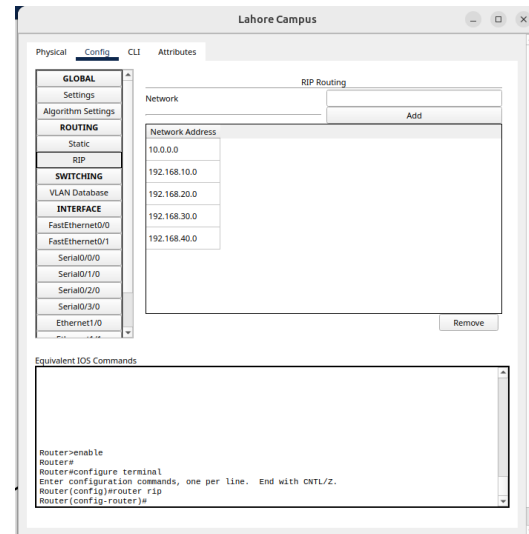
Routing is the process of transferring data between different networks, where dynamic routing enables routers to automatically exchange routing information without requiring manual configuration. This approach allows the network to adapt efficiently as routes are discovered and updated in real time.

In this project, the Routing Information Protocol (RIP) is implemented as the dynamic routing protocol to interconnect all campuses. It enables routers to share routing information automatically, discover remote networks, update routing tables dynamically, and maintain continuous communication between campuses. All routers are connected through serial WAN links and exchange network information to learn paths to different campuses. The RIP configuration includes network addresses such as 10.0.0.0, 192.168.10.0, 192.168.20.0,

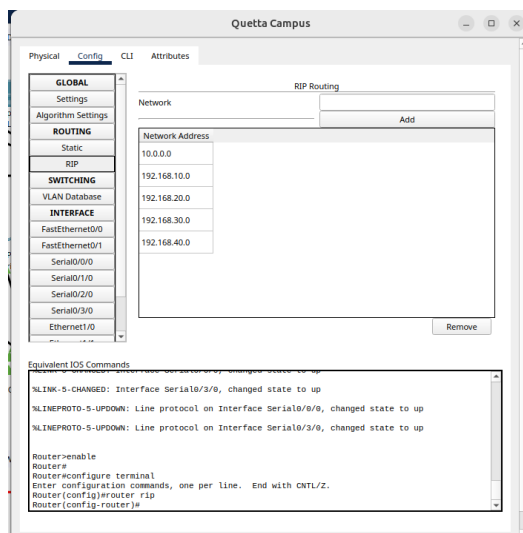
192.168.30.0, and 192.168.40.0. This implementation simplifies network management while ensuring reliable communication across all campuses.



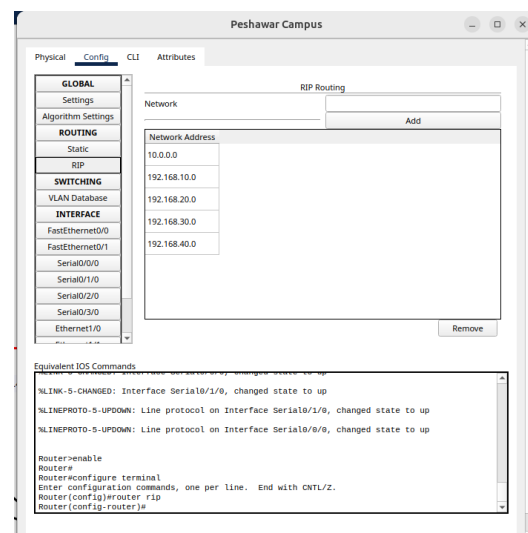
(A)



(B)



(C)



(D)

Figure 03: (A) Karachi Campus RIP Configuration, (B) Lahore Campus RIP Configuration, (C) Quetta Campus RIP Configuration, (D) Peshawar Campus RIP Configuration

3 Network Requirements Analysis

Before designing the network, a detailed requirement analysis was conducted to understand the overall structure and functional needs of the system, including the number of campuses, departments, end devices, and networking components required to ensure efficient communication. This planning phase also supported the selection of appropriate IP addressing schemes, subnetting strategies, and routing methods to maintain a well-organized and scalable

network design, and the project was ultimately developed to simulate a real-world university networking environment where multiple campuses and departments can communicate seamlessly with each other in a structured and efficient manner.

3.1 Number of Campuses

The network consists of four university campuses interconnected through routers using WAN serial links, where each campus represents a distinct network location in a different city, including Karachi, Lahore, Quetta, and Peshawar. Within each campus, multiple departments are connected through switches to ensure structured internal communication, while all campuses are integrated using Router-2811 configured with the RIP dynamic routing protocol. This interconnection enables seamless communication and efficient resource sharing between different university locations, forming a unified and scalable network architecture.

3.2 Number of Departments

Each campus is structured into five departments, where each department is assigned a separate switch and subnet to improve network organization, performance, and communication efficiency. The departments include Administration, Library, Examination, Accounts, and Education, and each department contains essential networking devices such as PCs, laptops, printers, and CCTV cameras. In addition, the Lahore and Quetta campuses include wireless access points along with wireless PCs to support WiFi-based communication. This segmented design improves overall network management by isolating traffic within departments and reducing unnecessary data flow across the network.

3.3 Hardware Requirements

Different networking hardware devices are used to build and simulate the university network. These devices help establish communication between campuses and departments.

Table 01: Hardware Devices Used in the Network Project and their Purpose

Device	Purpose
Router-2811	Used to connect campuses and perform routing
Switches-2950T-24	Used to connect departmental devices
PCs	Used for user communication and testing
Laptops	Used for communication inside departments
Printers	Used for network printing
CCTV Cameras	Used for monitoring and security
PT Access Points	Used for wireless communication
Fast Ethernet (Fa) Cables	Used for LAN connections
Ethernet (Eth) Cables	Used for device communication in LAN
Serial Cables	Used for router-to-router WAN connections

These hardware devices help create a complete university networking environment in Cisco Packet Tracer.

3.4 Software Requirements

Table 02: Software Devices Used in the Network Project and their Purpose

Software	Purpose
Cisco Packet Tracer	Used for network design, configuration, and simulation
Operating System	Used to run Cisco Packet Tracer
RIP Routing Protocol	Used for dynamic routing between campuses

Cisco Packet Tracer was the main software used in this project. It helped in creating the network topology, configuring devices, assigning IP addresses, and testing communication between all campuses and departments.

4 Base Network Design

The base network design forms a fundamental part of this project as it defines the overall structure and connectivity of the university network. A well-planned design improves communication efficiency, simplifies network management, and supports scalability and performance across the entire system. In this project, distinct IP ranges are assigned to each campus, and subnetting is used to divide the network into smaller departmental segments for better organization and control. WAN connections are implemented between routers to enable communication between all campuses, ensuring a fully interconnected and functional network infrastructure.

4.1 Selected Base Network IP

Private Class C IP addresses were selected for the university network. Different network ranges were assigned to each campus to keep the network organized and easy to manage.

Table 03: Base Network IP Address Allocation

Campus	Base Network IP
Karachi Campus	192.168.10.0
Lahore Campus	192.168.20.0
Quetta Campus	192.168.30.0
Peshawar Campus	192.168.40.0

For router-to-router WAN communication the network range was used: 10.0.0.0/30

These IP ranges helped separate campus networks and allowed proper routing between them.

4.2 Local Area Network Subnet Mask

For departmental Local Area Networks (LANs), a /28 subnet mask (255.255.255.240) is used, as each department requires a relatively small number of devices, including PCs, laptops, printers, CCTV cameras, and access points. This subnet size provides a total of 16 IP addresses, out of which 14 are usable host addresses, while one address is reserved for the network and

one for the broadcast address. Each department is assigned a separate subnet to improve network organization, control, and communication efficiency. For example, subnets such as 192.168.10.0/28, 192.168.10.16/28, and 192.168.10.32/28 are used within the network design. This approach helps reduce unnecessary traffic and simplifies overall network management.

4.3 Wide Area Network Subnet Mask

For router-to-router WAN communication, a /30 subnet mask (255.255.255.252) is used, as it is specifically designed for point-to-point links where only two router interfaces require IP addresses. This makes it an efficient choice for inter-router communication because it minimizes IP address wastage while maintaining proper connectivity between devices. A /30 subnet provides a total of four IP addresses, out of which two are usable host addresses, while one is reserved for the network and one for the broadcast address. In this project, the WAN connections are established between the Karachi, Lahore, Quetta, and Peshawar campus routers, and the use of /30 subnetting ensures efficient IP utilization and optimized WAN performance across the network.

4.4 Need for Subnetting

Subnetting is an essential part of this project as it divides a large network into smaller, more manageable subnetworks, where each department is assigned its own subnet to improve organization and communication efficiency. It offers several benefits, including improved network management, reduced traffic congestion, enhanced performance, better security, and more efficient utilization of IP addresses. Without subnetting, all devices would exist within a single large network, leading to increased traffic, poor performance, and difficulty in troubleshooting. In this project, /28 subnetting is used for departmental Local Area Networks (LANs), while /30 subnetting is implemented for WAN router links, resulting in a structured, scalable, and professionally organized network design.

4.5 Wide Area Network Connectivity Between Campuses

All campuses in the university network are connected through WAN serial links using Router-2811. Serial cables were used to establish communication between routers.

Table 04: WAN Connectivity and IP Addressing Scheme

Connected Campuses	WAN IP Addresses
Karachi ↔ Lahore	10.0.0.9 and 10.0.0.10
Karachi ↔ Quetta	10.0.0.1 and 10.0.0.2
Lahore ↔ Peshawar	10.0.0.13 and 10.0.0.14
Quetta ↔ Peshawar	10.0.0.5 and 10.0.0.6

RIP is configured as the dynamic routing protocol on all routers to enable automatic exchange of routing information between campuses. This allows routers to discover and update network paths without manual intervention, ensuring efficient and continuous communication across the entire network infrastructure. The RIP configuration includes the networks 10.0.0.0, 192.168.10.0, 192.168.20.0, 192.168.30.0, and 192.168.40.0, which collectively represent all campus and departmental segments. This WAN design ensures successful connectivity

between all campuses and departments, resulting in a fully integrated and operational university network.

5 Subnetting Design

Subnetting is a fundamental aspect of network design used to divide a large network into smaller, more manageable subnetworks. In this project, subnetting is implemented to create separate network segments for each department as well as for router-to-router connections, which enhances network organization, improves communication efficiency, and ensures proper IP address management across the system. Two subnet masks are utilized in the design: /28 for departmental LAN networks and /30 for WAN point-to-point communication between routers, resulting in a structured and efficient network architecture.

5.1 Applied Local Area Network Subnet Mask (/28)

For departmental Local Area Networks (LANs), a /28 subnet mask (255.255.255.240) is used to divide the network into smaller subnetworks suitable for departments with a limited number of connected devices. Each department includes PCs, laptops, printers, CCTV cameras, and in selected campuses, access points for wireless connectivity. This subnet configuration provides a total of 16 IP addresses, including 14 usable host addresses, one network address, and one broadcast address. For example, subnets such as 192.168.10.0/28, 192.168.10.16/28, and 192.168.10.32/28 are assigned to different departments. This approach ensures that each department operates on a separate network segment, improving organization, efficiency, and overall network management.

5.2 Applied Wide Area Network Subnet Mask (/30)

For router-to-router WAN communication, a /30 subnet mask (255.255.255.252) is used to support point-to-point serial links between routers, where only two IP addresses are required for communication. This subnet configuration is specifically designed for such links as it minimizes IP address wastage while maintaining efficient connectivity. A /30 subnet provides a total of 4 IP addresses, including 2 usable host addresses, 1 network address, and 1 broadcast address. This WAN subnetting approach ensures efficient utilization of IP resources and enables reliable and optimized communication between routers.

5.3 Subnet Calculation

Subnet calculation was performed to divide the network into smaller departmental subnetworks in a structured and efficient manner. For LAN subnetting, a /28 subnet mask is used, which results in a block size of 16 IP addresses per subnet. This value can be mathematically verified using standard IPv4 subnetting principles, where the number of host bits is determined as $32 - 28 = 4$, and the total number of addresses is calculated as $2^4 = 16$. This confirms that each /28 subnet provides 16 IP addresses in total, making it suitable for departmental network segmentation and ensuring proper address allocation within the LAN design.

Table 05: LAN Subnet Calculation Using /28

Subnet Number	Network Address	Usable Hosts	Broadcast Address
1	192.168.10.0/28	192.168.10.1 – 192.168.10.14	192.168.10.15
2	192.168.10.16/28	192.168.10.17 – 192.168.10.30	192.168.10.31
3	192.168.10.32/28	192.168.10.33 – 192.168.10.46	192.168.10.47
4	192.168.10.48/28	192.168.10.49 – 192.168.10.62	192.168.10.63
5	192.168.10.64/28	192.168.10.65 – 192.168.10.78	192.168.10.79

For WAN subnetting, a /30 subnet mask is used, which results in a block size of 4 IP addresses per subnet. This can be verified using standard IPv4 subnetting calculations, where the number of host bits is determined as $32 - 30 = 2$, and the total number of addresses is calculated as $2^2 = 4$. This confirms that each /30 subnet provides 4 IP addresses in total, making it suitable for point-to-point WAN links between routers and ensuring efficient use of IP address space.

Table 06: WAN Subnet Calculation Using /30

WAN Network	Router IPs	Broadcast Address
10.0.0.0/30	10.0.0.1 – 10.0.0.2	10.0.0.3
10.0.0.4/30	10.0.0.5 – 10.0.0.6	10.0.0.7
10.0.0.8/30	10.0.0.9 – 10.0.0.10	10.0.0.11
10.0.0.12/30	10.0.0.13 – 10.0.0.14	10.0.0.15

5.4 Number of Subnets and Hosts

Different subnetworks are designed for both departmental LANs and WAN links to ensure proper segmentation and efficient network utilization. For LAN subnetting, each campus consists of five departments, including Administration, Library, Examination, Accounts, and Education. With four campuses in total, the network requires $5 \times 4 = 20$ LAN subnets. Each LAN subnet is designed to support up to 14 usable host addresses, making it suitable for departmental-level device allocation and communication.

For WAN connectivity, four point-to-point serial links are established between routers to interconnect all campuses, including Karachi to Lahore, Karachi to Quetta, Lahore to Peshawar, and Quetta to Peshawar. This results in a total of 4 WAN subnets, where each subnet supports 2 usable host addresses. This structured subnet allocation ensures efficient communication between campuses while maintaining optimal use of IP addressing resources across the entire network.

5.5 IP Address Allocation Table

Table 07: Campus Wise IP Address Allocation Table

Campus	Department	Network Address	Default Gateway
Karachi	Administration	192.168.10.0/28	192.168.10.1
	Library	192.168.10.16/28	192.168.10.17
	Examination	192.168.10.32/28	192.168.10.33
	Accounts	192.168.10.48/28	192.168.10.49

Lahore	Education	192.168.10.64/28	192.168.10.65
	Administration	192.168.20.0/28	192.168.20.1
	Library	192.168.20.16/28	192.168.20.17
	Examination	192.168.20.32/28	192.168.20.33
	Accounts	192.168.20.48/28	192.168.20.49
Quetta	Education	192.168.20.64/28	192.168.20.65
	Administration	192.168.30.0/28	192.168.30.1
	Library	192.168.30.16/28	192.168.30.17
	Examination	192.168.30.32/28	192.168.30.33
	Accounts	192.168.30.48/28	192.168.30.49
Peshawar	Education	192.168.30.64/28	192.168.30.65
	Administration	192.168.40.0/28	192.168.40.1
	Library	192.168.40.16/28	192.168.40.17
	Examination	192.168.40.32/28	192.168.40.33
	Accounts	192.168.40.48/28	192.168.40.49
	Education	192.168.40.64/28	192.168.40.65

6 Campus Wise Network Design

The university network is divided into four campuses located in Karachi, Lahore, Quetta, and Peshawar, where each campus is interconnected through WAN serial links using Router-2811 to enable reliable inter-campus communication, while separate subnetworks are assigned to every department to improve network organization, management, and communication efficiency. Each campus consists of five departments connected through switches, and various devices such as PCs, laptops, printers, CCTV cameras, and access points are configured according to specific network requirements, resulting in a structured and scalable design that supports both internal departmental communication and inter-campus connectivity.

6.1 Karachi Campus

The Karachi campus is designed using a Router-2811 connected to five departmental switches, where each department is assigned a separate /28 subnet to ensure proper network segmentation and efficient communication. The campus includes the Administration, Library, Examination, Accounts, and Education departments, and each department is structured with essential network devices such as one PC, one laptop, one printer, and one CCTV camera. This design ensures organized traffic flow within the campus while maintaining clear separation between departmental networks. The Karachi campus router was connected with Lahore and Quetta campuses through serial WAN links.

Table 08: Karachi Campus Network Design

Department	Network Address	Default Gateway
Administration	192.168.10.0/28	192.168.10.1
Library	192.168.10.16/28	192.168.10.17
Examination	192.168.10.32/28	192.168.10.33
Accounts	192.168.10.48/28	192.168.10.49

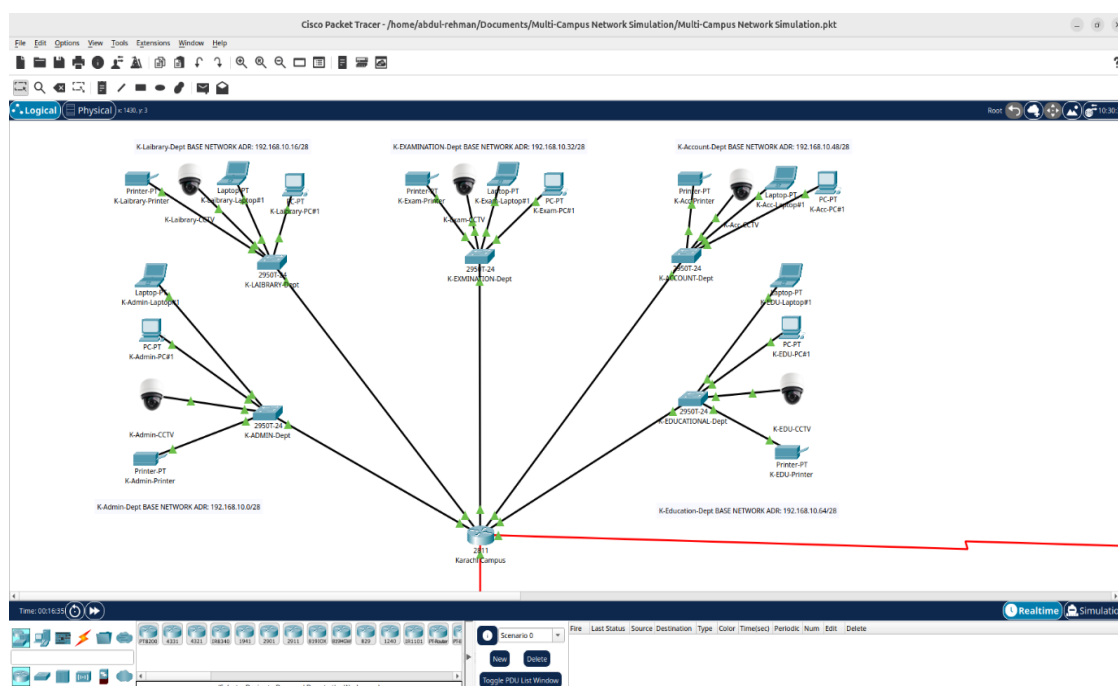


Figure 04: Karachi Campus LAN

6.2 Lahore Campus

The Lahore campus is implemented using a Router-2811 connected to five departmental switches, where each department is assigned a separate /28 subnet to ensure proper segmentation and efficient network management. The campus includes the Administration, Library, Examination, Accounts, and Education departments, and each department is configured with two PCs, one laptop, one printer, one CCTV camera, and one access point. One PC in each department is connected wirelessly through the access point, with WiFi password protection enabled to ensure secure communication. The Lahore campus router is also interconnected with the Karachi and Peshawar campuses through WAN serial links, enabling reliable inter-campus communication within the overall university network.

Table 09: Lahore Campus Network Design

Department	Network Address	Default Gateway
Administration	192.168.20.0/28	192.168.20.1
Library	192.168.20.16/28	192.168.20.17
Examination	192.168.20.32/28	192.168.20.33
Accounts	192.168.20.48/28	192.168.20.49
Education	192.168.20.64/28	192.168.20.65

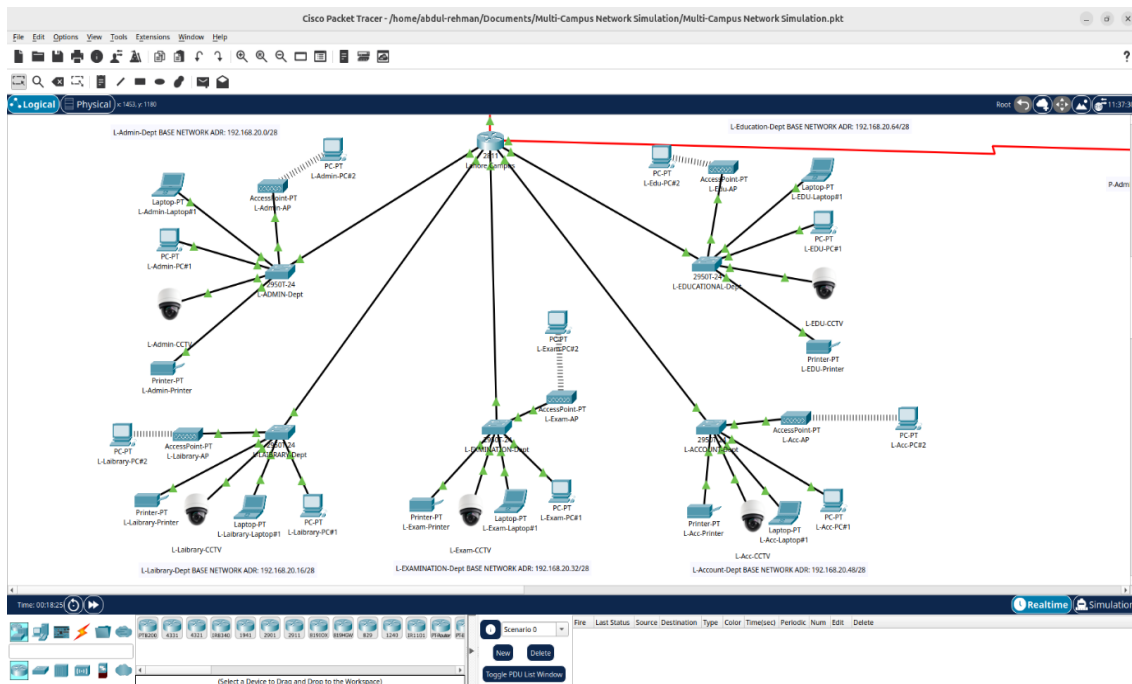


Figure 05: Lahore Campus LAN

6.3 Quetta Campus

The Quetta campus network is designed using a Router-2811 connected to five departmental switches, where each department is assigned a separate subnet to ensure proper segmentation and efficient network operation. The campus includes the Administration, Library, Examination, Accounts, and Education departments, and each department is configured with two PCs, one laptop, one printer, one CCTV camera, and one access point. Wireless communication is implemented through PT access points, where wireless PCs connect using secured WiFi passwords to maintain controlled access. The Quetta campus router is also connected with the Karachi and Peshawar campuses through serial WAN links, enabling smooth inter-campus communication within the university network.

Table 10: Quetta Campus Network Design

Department	Network Address	Default Gateway
Administration	192.168.30.0/28	192.168.30.1
Library	192.168.30.16/28	192.168.30.17
Examination	192.168.30.32/28	192.168.30.33
Accounts	192.168.30.48/28	192.168.30.49
Education	192.168.30.64/28	192.168.30.65

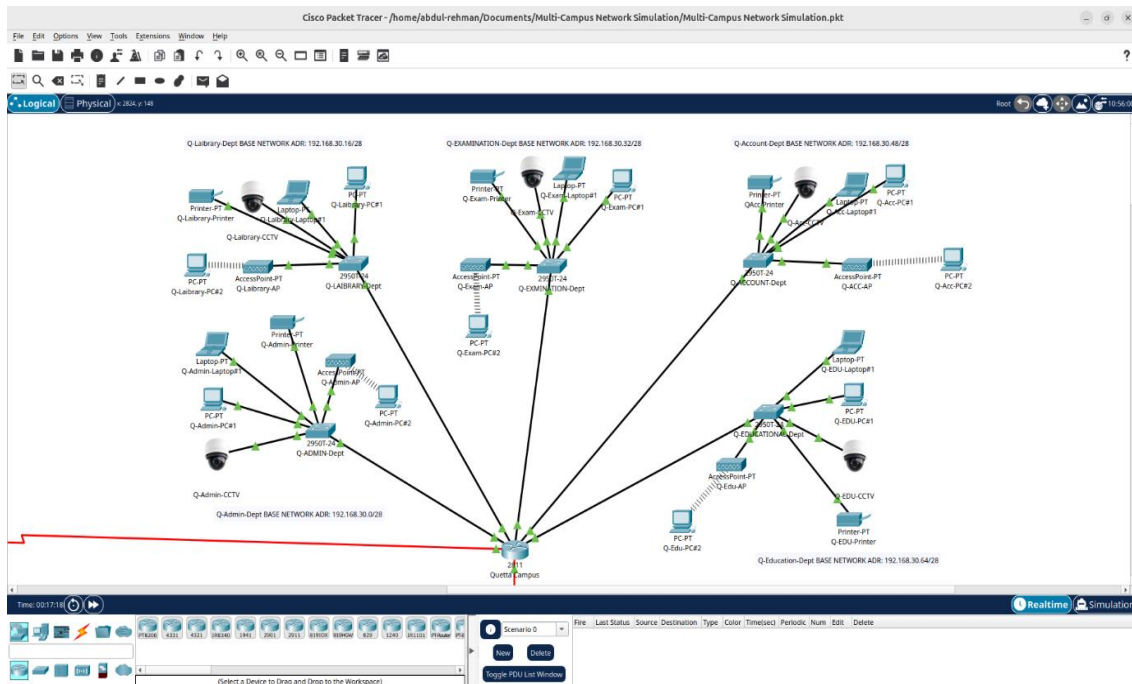


Figure 06: Quetta Campus LAN

6.4 Peshawar Campus

The Peshawar campus is implemented using a Router-2811 connected to five departmental switches, where each department is assigned a separate /28 subnet to ensure proper segmentation and efficient network management. The campus includes the Administration, Library, Examination, Accounts, and Education departments, and each department is configured with one PC, one laptop, one printer, and one CCTV camera. The Peshawar campus router is interconnected with the Lahore and Quetta campuses through WAN serial links, enabling seamless communication across the entire university network.

Table 11: Peshawar Campus Network Design

Department	Network Address	Default Gateway
Administration	192.168.40.0/28	192.168.40.1
Library	192.168.40.16/28	192.168.40.17
Examination	192.168.40.32/28	192.168.40.33
Accounts	192.168.40.48/28	192.168.40.49
Education	192.168.40.64/28	192.168.40.65

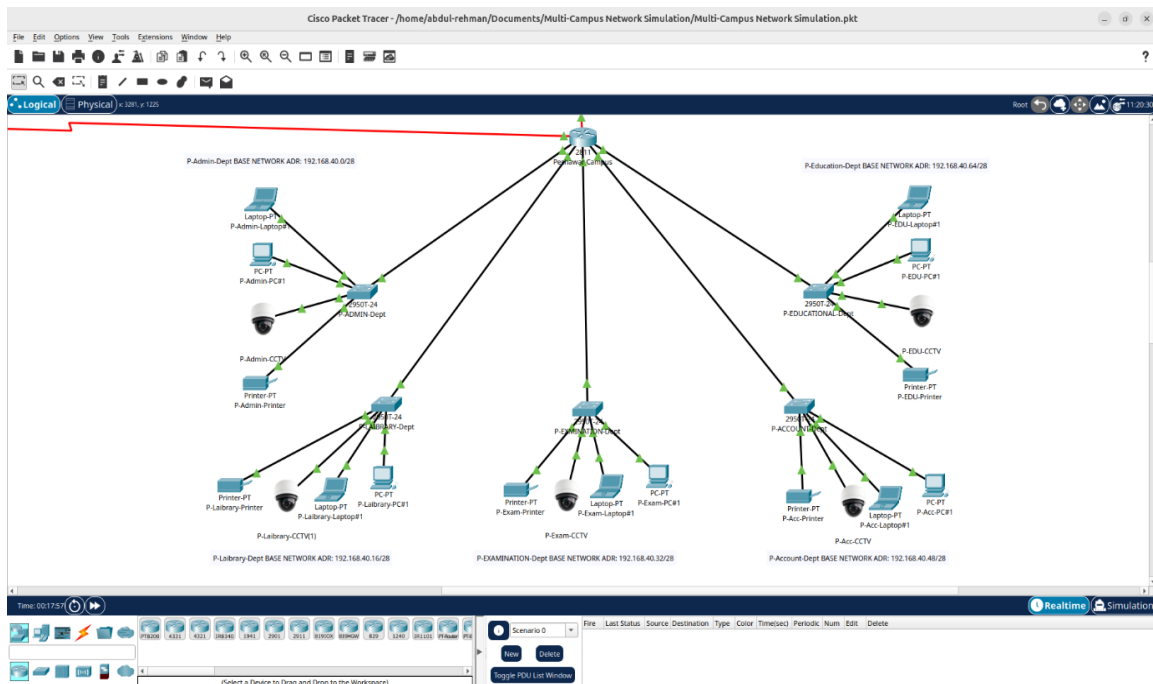


Figure 07: Peshawar Campus LAN

7 Department Wise Network Design

The university network is structured into multiple departments to enhance communication, improve management, and ensure proper network organization across all campuses, where each department is connected through a dedicated switch and assigned an individual /28 subnet for clear segmentation and efficient traffic handling, while every department includes essential networking devices such as PCs, laptops, printers, and CCTV cameras, and wireless access points are also deployed in selected campuses to support WiFi connectivity, resulting in an organized, scalable, and efficient networking environment that supports both wired and wireless communication across the entire university system.

7.1 Administration Department

The Administration Department is designed to manage administrative communication and operational activities within the university network, where a separate subnet is assigned to the department in each campus to ensure proper segmentation and efficient communication. The department includes PCs, laptops, printers, and CCTV cameras, while wireless PCs are also connected through access points in the Lahore and Quetta campuses. The following table presents the complete network configuration of the Administration Department across all campuses, including department name, network address, router interface, device names, assigned IP addresses, subnet masks, and default gateway information used in the overall network design.

Table 12: Administration Department IP Configuration in all Campuses

Karachi Campus Configuration	
Department Name	Administration

Network Address	192.168.10.0/28		
Router Interface	Fast Ethernet 0/0		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.10.2	255.255.255.240	192.168.10.1
Laptop#01	192.168.10.4	255.255.255.240	192.168.10.1
Printer	192.168.10.6	255.255.255.240	192.168.10.1
CCTV	192.168.10.7	255.255.255.240	192.168.10.1
Lahore Campus Configuration			
Department Name	Administration		
Network Address	192.168.20.0/28		
Router Interface	Fast Ethernet 0/0		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.20.2	255.255.255.240	192.168.20.1
PC#02 (Wireless)	192.168.20.3	255.255.255.240	192.168.20.1
Laptop	192.168.20.4	255.255.255.240	192.168.20.1
Printer	192.168.20.6	255.255.255.240	192.168.20.1
CCTV	192.168.20.7	255.255.255.240	192.168.20.1
Quetta Campus Configuration			
Department Name	Administration		
Network Address	192.168.30.0/28		
Router Interface	Fast Ethernet 0/0		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.30.2	255.255.255.240	192.168.30.1
PC#02 (Wireless)	192.168.30.3	255.255.255.240	192.168.30.1
Laptop	192.168.30.4	255.255.255.240	192.168.30.1
Printer	192.168.30.6	255.255.255.240	192.168.30.1
CCTV	192.168.30.7	255.255.255.240	192.168.30.1
Peshawar Campus Configuration			
Department Name	Administration		
Network Address	192.168.40.0/28		
Router Interface	Fast Ethernet 0/0		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.40.2	255.255.255.240	192.168.40.1
Laptop#01	192.168.40.4	255.255.255.240	192.168.40.1
Printer	192.168.40.6	255.255.255.240	192.168.40.1
CCTV	192.168.40.7	255.255.255.240	192.168.40.1

7.2 Library Department

The Library Department is designed to facilitate communication and resource sharing within the university library environment, where separate subnetworks are assigned in each campus to ensure proper organization and efficient network management. The department includes PCs, laptops, printers, and CCTV cameras, while wireless communication is also implemented in the Lahore and Quetta campuses through access points. The following table presents the complete network configuration of the Library Department across all campuses, including

department name, network address, router interface, device names, assigned IP addresses, subnet masks, and default gateway information used in the university network design.

Table 13: Library Department IP Configuration in all Campuses

Karachi Campus Configuration			
Department Name	Library		
Network Address	192.168.10.16/28		
Router Interface	Fast Ethernet 0/1		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.10.18	255.255.255.240	192.168.10.17
Laptop#01	192.168.10.20	255.255.255.240	192.168.10.17
Printer	192.168.10.22	255.255.255.240	192.168.10.17
CCTV	192.168.10.23	255.255.255.240	192.168.10.17
Lahore Campus Configuration			
Department Name	Library		
Network Address	192.168.20.16/28		
Router Interface	Fast Ethernet 0/1		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.20.18	255.255.255.240	192.168.20.17
PC#02 (Wireless)	192.168.20.19	255.255.255.240	192.168.20.17
Laptop	192.168.20.20	255.255.255.240	192.168.20.17
Printer	192.168.20.22	255.255.255.240	192.168.20.17
CCTV	192.168.20.23	255.255.255.240	192.168.20.17
Quetta Campus Configuration			
Department Name	Library		
Network Address	192.168.30.16/28		
Router Interface	Fast Ethernet 0/1		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.30.18	255.255.255.240	192.168.30.17
PC#02 (Wireless)	192.168.30.19	255.255.255.240	192.168.30.17
Laptop	192.168.30.20	255.255.255.240	192.168.30.17
Printer	192.168.30.22	255.255.255.240	192.168.30.17
CCTV	192.168.30.23	255.255.255.240	192.168.30.17
Peshawar Campus Configuration			
Department Name	Library		
Network Address	192.168.40.16/28		
Router Interface	Fast Ethernet 0/1		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.40.18	255.255.255.240	192.168.40.17
Laptop#01	192.168.40.20	255.255.255.240	192.168.40.17
Printer	192.168.40.22	255.255.255.240	192.168.40.17
CCTV	192.168.40.23	255.255.255.240	192.168.40.17

7.3 Examination Department

The Examination Department is designed to manage examination-related communication and data sharing between devices, where separate subnetworks are assigned in each campus to reduce unnecessary network traffic and ensure proper organization. The department includes PCs, laptops, printers, and CCTV cameras, while in the Lahore and Quetta campuses, wireless PCs are also connected through access points secured with WiFi passwords. The following table presents the complete network configuration of the Examination Department across all campuses, including department name, network address, router interface, device names, assigned IP addresses, subnet masks, and default gateway information used in the university network design.

Table 14: Examination Department IP Configuration in all Campuses

Karachi Campus Configuration			
Department Name	Examination		
Network Address	192.168.10.32/28		
Router Interface	Ethernet 1/0		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.10.34	255.255.255.240	192.168.10.33
Laptop#01	192.168.10.36	255.255.255.240	192.168.10.33
Printer	192.168.10.38	255.255.255.240	192.168.10.33
CCTV	192.168.10.39	255.255.255.240	192.168.10.33
Lahore Campus Configuration			
Department Name	Examination		
Network Address	192.168.20.32/28		
Router Interface	Ethernet 1/0		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.20.34	255.255.255.240	192.168.20.33
PC#02 (Wireless)	192.168.20.35	255.255.255.240	192.168.20.33
Laptop	192.168.20.36	255.255.255.240	192.168.20.33
Printer	192.168.20.38	255.255.255.240	192.168.20.33
CCTV	192.168.20.39	255.255.255.240	192.168.20.33
Quetta Campus Configuration			
Department Name	Examination		
Network Address	192.168.30.32/28		
Router Interface	Ethernet 1/0		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.30.34	255.255.255.240	192.168.30.33
PC#02 (Wireless)	192.168.30.35	255.255.255.240	192.168.30.33
Laptop	192.168.30.36	255.255.255.240	192.168.30.33
Printer	192.168.30.38	255.255.255.240	192.168.30.33
CCTV	192.168.30.39	255.255.255.240	192.168.30.33
Peshawar Campus Configuration			
Department Name	Examination		

Network Address	192.168.40.32/28		
Router Interface	Ethernet 1/0		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.40.34	255.255.255.240	192.168.40.33
Laptop#01	192.168.40.36	255.255.255.240	192.168.40.33
Printer	192.168.40.38	255.255.255.240	192.168.40.33
CCTV	192.168.40.39	255.255.255.240	192.168.40.33

7.4 Accounts Department

The Accounts Department is implemented to support financial and accounting communication within the university network, where separate subnetworks are assigned in each campus to enhance security, organization, and management of departmental data. The department includes PCs, laptops, printers, and CCTV cameras, while wireless communication is also available in the Lahore and Quetta campuses through PT access points to support secure connectivity. The following table presents the complete network configuration of the Accounts Department across all campuses, including department name, network address, router interface, device names, assigned IP addresses, subnet masks, and default gateway information used in the university network design.

Table 15: Accounts Department IP Configuration in all Campuses

Karachi Campus Configuration			
Department Name	Accounts		
Network Address	192.168.10.48/28		
Router Interface	Ethernet 1/1		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.10.50	255.255.255.240	192.168.10.49
Laptop#01	192.168.10.52	255.255.255.240	192.168.10.49
Printer	192.168.10.54	255.255.255.240	192.168.10.49
CCTV	192.168.10.55	255.255.255.240	192.168.10.49
Lahore Campus Configuration			
Department Name	Accounts		
Network Address	192.168.20.48/28		
Router Interface	Ethernet 1/1		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.20.50	255.255.255.240	192.168.20.49
PC#02 (Wireless)	192.168.20.51	255.255.255.240	192.168.20.49
Laptop	192.168.20.52	255.255.255.240	192.168.20.49
Printer	192.168.20.54	255.255.255.240	192.168.20.49
CCTV	192.168.20.55	255.255.255.240	192.168.20.49
Quetta Campus Configuration			
Department Name	Accounts		
Network Address	192.168.30.48/28		
Router Interface	Ethernet 1/1		
Device Name	IP Address	Subnet Mask	Default Gateway

PC#01	192.168.30.50	255.255.255.240	192.168.30.49
PC#02 (Wireless)	192.168.30.51	255.255.255.240	192.168.30.49
Laptop	192.168.30.52	255.255.255.240	192.168.30.49
Printer	192.168.30.54	255.255.255.240	192.168.30.49
CCTV	192.168.30.55	255.255.255.240	192.168.30.49
Peshawar Campus Configuration			
Department Name	Accounts		
Network Address	192.168.40.48/28		
Router Interface	Ethernet 1/1		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.40.50	255.255.255.240	192.168.40.49
Laptop#01	192.168.40.52	255.255.255.240	192.168.40.49
Printer	192.168.40.54	255.255.255.240	192.168.40.49
CCTV	192.168.40.55	255.255.255.240	192.168.40.49

7.5 Education Department

The Education Department is designed to facilitate educational communication and data sharing between devices within the university network, where structured subnetting is applied in each campus to ensure proper organization and efficient network management. The department includes PCs, laptops, printers, and CCTV cameras, while in the Lahore and Quetta campuses, wireless PCs are also connected through access points secured with password protection. The following table presents the complete network configuration of the Education Department across all campuses, including department name, network address, router interface, device names, assigned IP addresses, subnet masks, and default gateway information used in the university network design.

Table 16: Education Department IP Configuration in all Campuses

Karachi Campus Configuration			
Department Name	Education		
Network Address	192.168.10.64/28		
Router Interface	Ethernet 1/2		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.10.66	255.255.255.240	192.168.10.65
Laptop#01	192.168.10.68	255.255.255.240	192.168.10.65
Printer	192.168.10.70	255.255.255.240	192.168.10.65
CCTV	192.168.10.71	255.255.255.240	192.168.10.65
Lahore Campus Configuration			
Department Name	Education		
Network Address	192.168.20.64/28		
Router Interface	Ethernet 1/2		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.20.66	255.255.255.240	192.168.20.65
PC#02 (Wireless)	192.168.20.67	255.255.255.240	192.168.20.65

Laptop	192.168.20.68	255.255.255.240	192.168.20.65
Printer	192.168.20.70	255.255.255.240	192.168.20.65
CCTV	192.168.20.71	255.255.255.240	192.168.20.65
Quetta Campus Configuration			
Department Name	Education		
Network Address	192.168.30.64/28		
Router Interface	Ethernet 1/2		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.30.66	255.255.255.240	192.168.30.65
PC#02 (Wireless)	192.168.30.67	255.255.255.240	192.168.30.65
Laptop	192.168.30.68	255.255.255.240	192.168.30.65
Printer	192.168.30.70	255.255.255.240	192.168.30.65
CCTV	192.168.30.71	255.255.255.240	192.168.30.65
Peshawar Campus Configuration			
Department Name	Education		
Network Address	192.168.40.64/28		
Router Interface	Ethernet 1/2		
Device Name	IP Address	Subnet Mask	Default Gateway
PC#01	192.168.40.66	255.255.255.240	192.168.40.65
Laptop#01	192.168.40.68	255.255.255.240	192.168.40.65
Printer	192.168.40.70	255.255.255.240	192.168.40.65
CCTV	192.168.40.71	255.255.255.240	192.168.40.65

8 Network Devices and Their Roles

8.1 Router-2811

A router is a networking device used to connect multiple networks and enable communication between them by forwarding data packets toward their destination IP addresses. It operates primarily using IP-based communication and the TCP/IP protocol suite, and is widely used in WAN environments and internet connectivity to manage inter-network data transfer efficiently. In this project, Router-2811 are used to interconnect all university campuses through serial WAN links, while the RIP dynamic routing protocol is configured on each router to automatically exchange routing information and maintain updated routing tables for optimal path selection. The router examines incoming packets, determines the destination network, and forwards them through the most appropriate route based on routing table decisions.

In the implemented network, routers play a central role by connecting all campuses, enabling dynamic routing through RIP, managing communication between different networks, and linking LAN and WAN segments into a unified system. They ensure reliable data forwarding between campuses by continuously updating routing information and selecting efficient paths for packet delivery across the entire university network.



Figure 08

8.2 Switch-2950T-24

A switch is a networking device used to connect multiple devices within the same LAN, enabling efficient communication between devices such as PCs, laptops, printers, and CCTV cameras. It operates primarily using Ethernet standards and MAC address-based forwarding to ensure accurate data delivery within a local network segment. In this project, Switches-2950T-24 are deployed in each department to connect end devices and facilitate communication within the assigned subnet. The switch processes incoming data frames by identifying the destination MAC address and forwarding them only to the intended device, which improves performance and reduces unnecessary network traffic.

In the implemented network, switches play a key role in connecting all departmental devices, managing LAN-level communication, reducing congestion, and improving overall efficiency. They also serve as an intermediate layer between end devices and routers, ensuring that local traffic is handled within the department while only necessary data is forwarded to other networks through the router.



Figure 09

8.3 Access Point

An access point is a wireless networking device that enables WiFi communication by allowing wireless devices to connect to a wired local area network. It operates using IEEE 802.11 standards, where IEEE (Institute of Electrical and Electronics Engineers) is an international organization that defines global networking and communication standards to ensure compatibility between different devices. In this project, PT access points are implemented in the Lahore and Quetta campuses, where wireless PCs connect through secured WiFi passwords for controlled access. The access point receives wireless signals from devices and bridges them with the wired network, enabling seamless communication between both network types.

In the implemented network, access points provide wireless connectivity within departments, support user mobility, and extend network access beyond wired infrastructure. They also ensure secure communication through password protection while integrating wireless devices

into the existing LAN environment, improving flexibility and overall usability of the university network.



Figure 10

8.4 Personal Computer (PC) and LAPTOPS

PCs and laptops are end-user devices used for communication, data sharing, and network testing, allowing users to access network resources and interact with other connected systems. These devices operate primarily using the TCP/IP protocol suite, IPv4 addressing, and both Ethernet and WiFi communication standards to enable reliable data exchange within a network environment. In this project, PCs and laptops are connected to switches through Ethernet links, while in the Lahore and Quetta campuses, some PCs also connect wirelessly through access points to support flexible communication. These devices use assigned IP addresses to send and receive data and communicate with both local and remote networks through routers.

In the implemented network, PCs and laptops play an essential role in enabling user communication, testing network connectivity, sharing data across campuses, accessing network resources, and supporting both wired and wireless communication within the university environment.



Figure 11

8.5 Printer

A printer is an output device used to produce hard copies of digital documents and reports, and when integrated into a network, it enables multiple users to access printing services efficiently. Network printers operate primarily using the TCP/IP protocol suite and standard network printing services to receive and process print requests from connected devices. In this project, printers are deployed in each department and connected through switches, where each printer

is assigned a static IP address to ensure consistent accessibility across the network. These printers receive print commands over the LAN and execute them by producing the required documents for users within the same network environment.

In the implemented system, printers play an important role by providing centralized printing services, supporting document sharing, facilitating departmental communication, and enabling efficient resource sharing within the university network.



Figure 12

8.6 Closed-Circuit Television (CCTV) Camera

A CCTV camera is a security and monitoring device used to observe activities within a network environment, contributing to improved safety and surveillance in organizational settings. It operates primarily through IP-based communication using the TCP/IP protocol suite, enabling real-time transmission of video data over the network. In this project, CCTV cameras are installed in each department and connected through switches, where each device is assigned a static IP address to ensure consistent access and reliable monitoring. These cameras continuously transmit video streams across the LAN, allowing authorized users to monitor departmental activities through the network infrastructure.

In the implemented system, CCTV cameras play a key role in enhancing security monitoring, improving campus-wide surveillance, supporting network-based video transmission, and ensuring overall safety within each department of the university environment.



Figure 13

9 Network Topology Design

Network topology refers to the physical and logical arrangement of devices within a network, illustrating how components such as routers, switches, PCs, printers, and access points are interconnected to enable communication. It defines the overall structure of data flow and connectivity within a system, ensuring that all network elements operate in a coordinated manner. In this project, both physical and logical topologies are implemented to design a multi-campus university network, enabling efficient communication across campuses and departments through a combination of wired and wireless connections.

The network architecture includes four campus routers, multiple departmental switches, end-user devices, wireless access points, and WAN serial links between campuses. This structured design ensures seamless connectivity and efficient data exchange between all network components, supporting reliable communication across the entire university system.

9.1 Physical Topology

Physical topology represents the actual physical arrangement of networking devices and the way they are interconnected within the network infrastructure. In this project, a star topology is implemented inside each campus, where all departmental devices are connected through centralized switches. PCs, laptops, printers, CCTV cameras, and access points are directly connected to their respective departmental switches, while each switch is connected to the campus router to establish communication between departments and external networks.

For inter-campus communication, Router-2811 devices are connected through serial WAN links using point-to-point connections between campuses including Karachi, Lahore, Quetta, and Peshawar. Ethernet and Fast Ethernet cables are used for LAN communication, while serial cables are used for WAN router-to-router connectivity. This star-based physical design provides better scalability, simplified troubleshooting, centralized management, and efficient communication throughout the university network infrastructure.

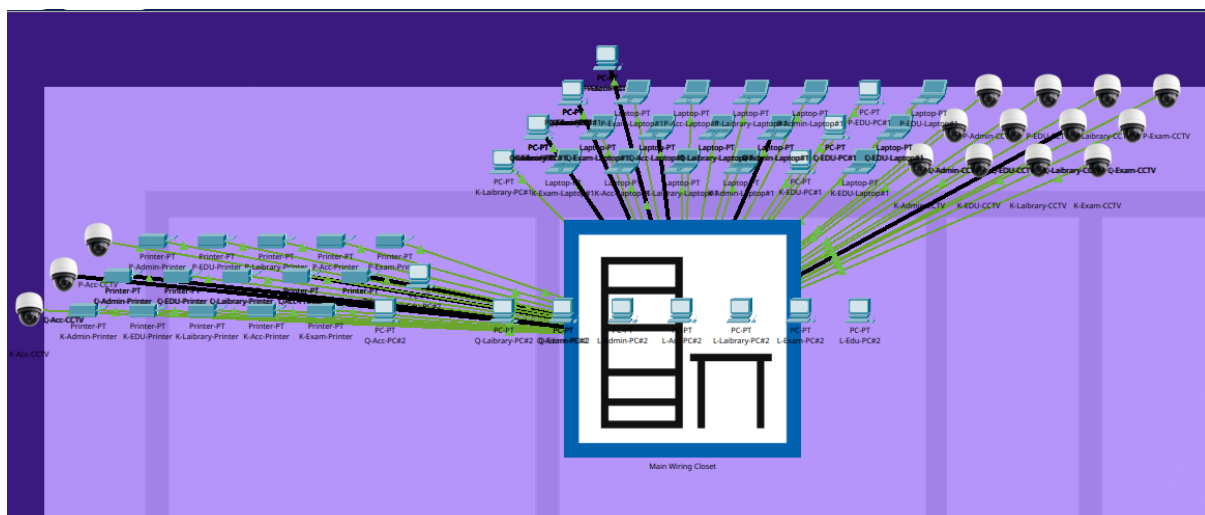


Figure 14: Physical Topology Network

9.2 Logical Topology

Dynamic Host Configuration Protocol (DHCP) is implemented within the university network to support automatic IP address assignment for connected devices. DHCP allows devices to obtain valid network configuration dynamically without requiring manual IP assignment, improving efficiency and reducing administrative workload.

The network design incorporates both static and dynamic IP addressing methods to maintain a balanced and flexible communication environment. Static addressing is applied to critical network infrastructure devices for reliable connectivity, while DHCP-based dynamic addressing is used for selected client devices to simplify network administration, reduce configuration errors, and improve scalability across the university network.

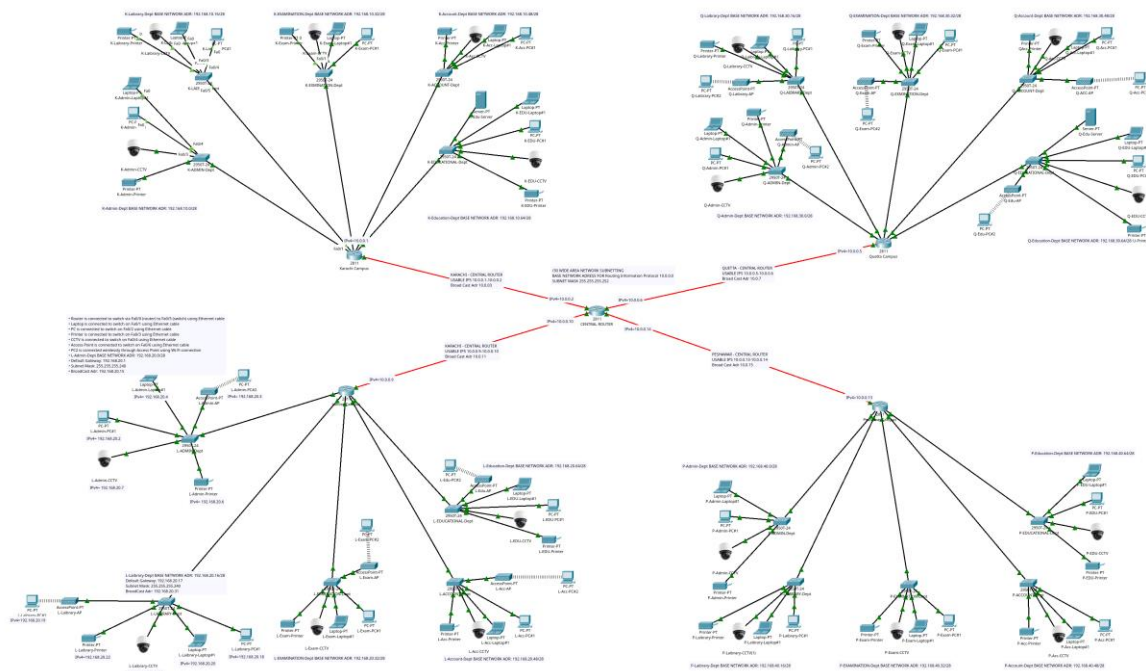


Figure 15: Logical Network Topology

9.3 Inter-Campus Connectivity

Inter-campus connectivity refers to the communication established between different university campuses through WAN links, enabling seamless data exchange across geographically separated locations. In this project, all campuses are interconnected using Router-2811 through serial WAN connections, where the RIP is implemented to automatically manage routing and ensure dynamic exchange of network information between routers. This allows each router to learn and update routes without manual configuration, maintaining continuous communication across the entire network.

As a result of this design, all campuses are able to communicate successfully with each other, allowing devices to exchange data across different locations while supporting both wired and wireless communication within the unified university network infrastructure.

Table 17: Campus Connections were Implemented

Connected Campuses	WAN IP Addresses
Karachi ↔ Central Device	10.0.0.1 and 10.0.0.2
Quetta ↔ Central Device	10.0.0.5 and 10.0.0.6
Lahore ↔ Central Device	10.0.0.9 and 10.0.0.10
Peshawar ↔ Central Device	10.0.0.13 and 10.0.0.14

10 Network Devices Configuration

After completing the network design, all networking devices were configured in Cisco Packet Tracer according to the project requirements to ensure proper communication between departments and campuses. Configuration settings were applied to routers, switches, access points, PCs, and wireless devices to establish a fully functional network environment. Static IP addressing was assigned to all end devices to maintain consistent identification, while RIP dynamic routing was implemented on routers to enable automatic communication between campuses. All configurations were carried out using the graphical interface of Cisco Packet Tracer, ensuring accurate setup and operational connectivity across the entire network.

10.1 Router Configuration

Routers (Model 2811) are configured to interconnect all university campuses and manage communication between different network segments within the system. IP addresses and subnet masks are assigned to Fast Ethernet, Ethernet, and serial interfaces based on their functional roles. Fast Ethernet and Ethernet interfaces are used for departmental LAN connectivity, while serial interfaces are configured for WAN communication between routers using /30 subnetting. Each departmental interface is assigned a default gateway corresponding to its subnet to ensure proper internal communication within each network segment.

After completing the IP configuration, all router interfaces are activated to enable full network connectivity. The routers successfully forward data packets between departments, campuses, wired devices, and wireless devices, ensuring seamless communication across the entire university network infrastructure.

10.2 Routing Information Protocol

Routing Information Protocol dynamic routing is configured on all routers through the routing configuration section in Cisco Packet Tracer to enable automatic exchange of routing information across the network. The configured RIP process includes network addresses 10.0.0.0, 192.168.10.0, 192.168.20.0, 192.168.30.0, and 192.168.40.0, which represent all major campus and departmental networks. Through RIP, routers automatically exchange routing updates, learn remote network routes, and maintain updated routing tables to ensure continuous communication between all campuses.

After RIP configuration, all routers successfully learned routes to remote networks, enabling smooth inter-campus communication and accurate packet delivery across the system. This dynamic routing approach simplifies network administration by reducing the need for manual route configuration while ensuring reliable and efficient connectivity throughout the university network.

10.3 Switch Configuration

Switches (Model 2950T-24) are used in all departments to connect networking devices within the local area network environment. These switches are connected to PCs, laptops, printers, CCTV cameras, and access points using Fast Ethernet and Ethernet cables to establish reliable LAN connectivity. The switching process is based on MAC address learning and forwarding, where each switch intelligently directs data frames only to the intended destination device, improving communication efficiency within the network.

All switches operate effectively to ensure proper communication between end devices and routers by maintaining organized traffic flow within each departmental network. This design supports stable LAN connectivity, reduces unnecessary data transmission, and enhances overall network performance across all departments.

10.4 Access Point Configuration

PT access points are configured in the Lahore and Quetta campuses to provide wireless communication within departmental networks. Each access point is set up with a wireless network name (SSID), WiFi password, and appropriate IP addressing parameters to ensure proper integration with the existing LAN infrastructure. Wireless PCs connect to these access points using WiFi connectivity, where authentication is enforced through password protection to allow only authorized devices to access the network.

The access points successfully bridge wireless devices with the wired LAN environment, enabling seamless communication between both network types while maintaining secure and controlled wireless access within the university network.

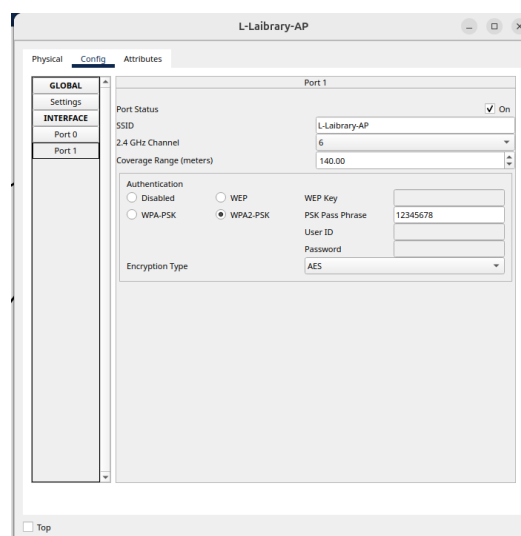


Figure 16: Access Point Wireless Configuration

10.5 Personal Computer (PC) Internet Protocol (IP) Configuration

All PCs and laptops are configured manually using static IP addressing to ensure consistent and controlled network communication. Each device is assigned an IP address, subnet mask, and default gateway based on the subnet design of its respective department, allowing proper integration within the network structure.

After configuration, all devices successfully communicate with other departmental devices, routers, and remote campuses. The use of static IP addressing helps maintain a stable and well-organized network environment, ensuring reliable communication and simplified network management across the entire university system.

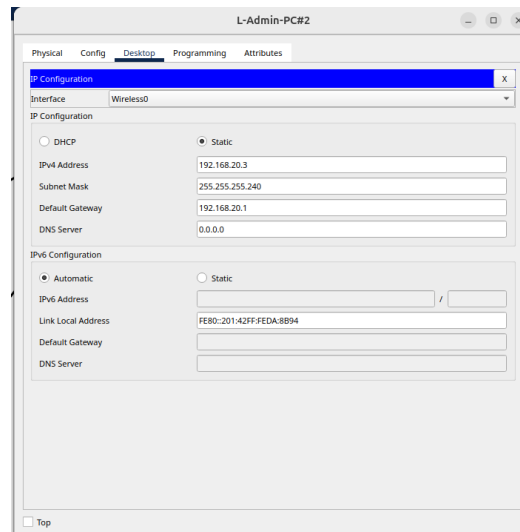


Figure 17: PC IP Configuration

10.6 Dynamic Host Configuration Protocol (DHCP) Configuration

Dynamic Host Configuration Protocol (DHCP) is configured in selected departments of the university network to automate the process of IP address allocation for end-user devices. The DHCP service automatically assigns IP addresses, subnet masks, and default gateway information from a predefined address pool whenever a device connects to the network. This reduces the need for manual IP configuration and improves the overall efficiency of network management.

In this project, both static and dynamic IP addressing techniques are implemented. Static IP addressing is used for routers, printers, CCTV cameras, and other important infrastructure devices to ensure stable and predictable communication, while DHCP-based dynamic addressing is used for selected end-user devices to simplify configuration and improve scalability within the network environment.

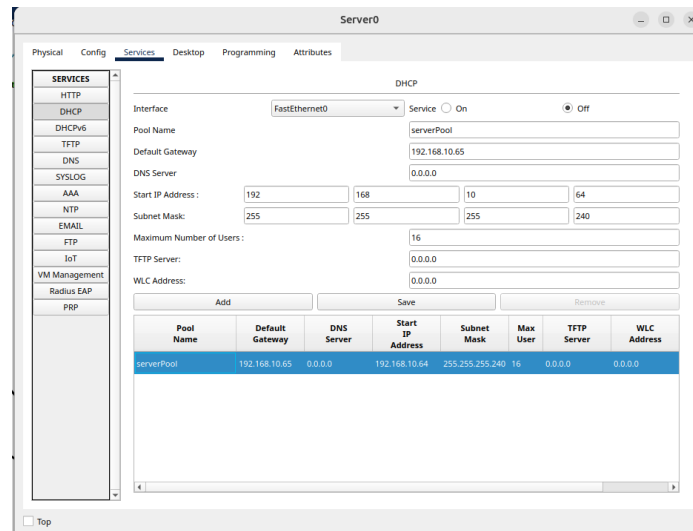


Figure 18: DHCP IP Configuration

10.7 Wireless Device Configuration

Wireless PCs in the Lahore and Quetta campuses are configured to communicate through PT access points to enable secure wireless connectivity within the network. Each wireless device is set up by selecting the appropriate wireless network, entering the WiFi password for authentication, and assigning a static IP address according to the departmental subnet design.

After successful configuration, wireless PCs communicate effectively with wired devices, other wireless devices, and remote campuses. Network connectivity is verified using the Simple Protocol Data Unit (PDU) tool in Cisco Packet Tracer, confirming that wireless communication is functioning correctly across the university network.

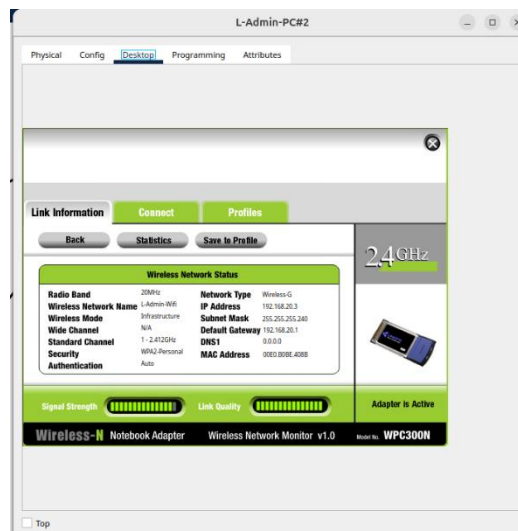


Figure 19: Wireless PC Connected with Access Point

11 Cisco Packet Tracer Implementation

Cisco Packet Tracer is used to design, configure, and simulate the complete university network project, where all networking devices, connections, IP addressing schemes, and routing

configurations are implemented within a controlled simulation environment. This allows accurate modelling of a real-world network infrastructure and validation of communication between different components.

The implementation process includes building the network topology, establishing device connections, assigning IP addresses, configuring routing protocols, and testing communication between campuses. The completed simulation successfully represents a real-world university networking environment, supporting both wired and wireless communication across the entire system.

11.1 Network Topology

The network topology is designed in Cisco Packet Tracer according to the project requirements to represent a structured multi-campus university network. The topology consists of four campuses interconnected through WAN serial links, where each campus is built around a Router-2811 connected to five 2950T-24 switches along with end devices such as PCs, laptops, printers, and CCTV cameras. In the Lahore and Quetta campuses, PT access points and wireless PCs are also included to support wireless communication in addition to the wired infrastructure. Each department is connected through a dedicated switch and assigned a separate subnet to ensure proper segmentation and efficient network management, while WAN connectivity between campuses is established through serial links.

This topology successfully supports communication at multiple levels, including inter-department communication within campuses, inter-campus communication across the WAN, and both wired and wireless data exchange throughout the network.

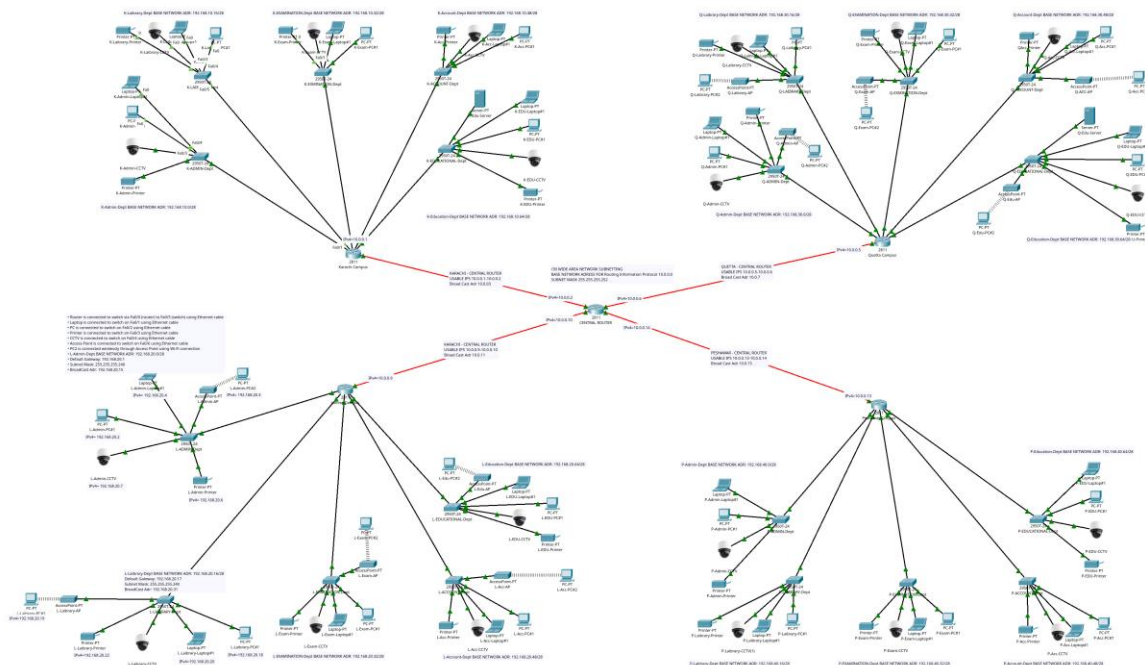


Figure 20: Complete University Network Topology in Cisco Packet Tracer

11.2 Device Naming Scheme

A proper naming scheme was used for all devices to make the network organized and easy to manage.

Routers were named according to campus locations:

- Karachi Router
- Lahore Router
- Quetta Router
- Peshawar Router

Switches were named according to departments and campuses, such as:

- K-Admin-SW
- L-Library-SW
- Q-Exam-SW
- P-Accounts-SW

End devices were also named according to their functions:

- PCs
- Laptops
- Printers
- CCTV Cameras
- Access Points

The naming scheme helped identify devices easily during configuration and testing.

11.3 IP Address Assignment

Static IP addressing was used throughout the project for all end devices and router interfaces. Different IP ranges were assigned to each campus:

- Karachi Campus → 192.168.10.0
- Lahore Campus → 192.168.20.0
- Quetta Campus → 192.168.30.0
- Peshawar Campus → 192.168.40.0

Departmental Local Area Networks (LANs) used: /28 = 255.255.255.240

WAN serial links between routers used: /30 = 255.255.255.252

Every device was configured with:

- IP Address
- Subnet Mask
- Default Gateway

The organized IP addressing scheme improved network communication and management.

11.4 Cabling and Connections

Different cable types were used in Cisco Packet Tracer to establish communication between devices.

The following cables were used:

- Fast Ethernet Cable → Used for LAN communication
- Ethernet Cable → Used to connect devices within departments
- Serial Cable → Used for router-to-router WAN communication

Fast Ethernet and Ethernet cables connected:

- Routers to switches
- Switches to PCs
- Switches to printers
- Switches to CCTV cameras
- Access points to switches

Serial cables connected routers between campuses for WAN communication. All cabling connections were implemented successfully and allowed proper data transfer between devices.

12 Connectivity Testing and Verification

After completing the network configuration, multiple connectivity tests are performed in Cisco Packet Tracer to verify communication between devices, departments, and campuses. The testing process confirms that all devices communicate successfully through the configured network, ensuring proper implementation of IP addressing, subnetting, and routing.

Cisco Packet Tracer simulation mode is used for validation, where the “Add Simple Protocol Data Unit” tool is utilized to send test packets between devices. This allows verification of end-to-end connectivity and confirms that data is being correctly transmitted across the entire university network.

12.1 Connectivity Testing Method

Connectivity testing is performed using the Simple Protocol Data Unit (PDU) tool available in Cisco Packet Tracer to verify successful packet delivery between devices across the network. This tool is used to check whether data packets can travel correctly from a source device to a destination device, ensuring proper network configuration and communication.

The testing procedure involves selecting the “Add Simple PDU” tool ( icon), choosing the source device, selecting the destination device, and observing the packet delivery status in the simulation environment. Successful packet delivery confirms correct communication between devices. This method is used to validate department-to-department communication, campus-to-campus communication, and both wired and wireless connectivity within the university network.

12.2 Inter Department Communication

Communication between different departments within the same campus is tested successfully to verify internal network functionality. Test scenarios include communication from the Administration Department to the Library Department, the Examination Department to the Accounts Department, and the Education Department to the Administration Department. In all cases, data packets are successfully exchanged through switches and routed appropriately within the campus network.

The successful delivery of packets confirms correct implementation of IP addressing, proper subnetting, functional switch-level communication, and correctly configured router interfaces, ensuring reliable and structured intra-campus communication across all departments.

12.3 Inter Campus Communication

Communication between different campuses is tested successfully using the Simple Protocol Data Unit tool to validate inter-campus connectivity across the WAN infrastructure. Test cases include communication from the Karachi campus to the Lahore and Quetta campuses, as well as communication from the Lahore campus to Peshawar, and from Quetta to Peshawar. In all scenarios, data packets are successfully delivered across routers through WAN serial links.

The successful packet transmission confirms that RIP dynamic routing is functioning correctly between all routers, enabling proper route discovery and data exchange. It also verifies that WAN serial link communication is stable and that inter-campus connectivity is fully operational across the entire university network.

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit	Delete
	Successful	L-Ed...	P-Admin-L...	ICMP		0.000	N	0	(edit)	(delete)
	Successful	K-Ac...	L-Edu-PC#2	ICMP		0.000	N	1	(edit)	(delete)
	Successful	K-Ex...	Q-Laibrary...	ICMP		0.000	N	2	(edit)	(delete)
	Successful	K-Ex...	Q-Admin...	ICMP		0.000	N	3	(edit)	(delete)
	Successful	Q-ED...	P-EDU-La...	ICMP		0.000	N	4	(edit)	(delete)

Figure 21: Inter Campus Communication

12.4 Wireless Connectivity Testing

Wireless communication is tested in the Lahore and Quetta campuses where PT access points are deployed to provide WiFi connectivity within departmental networks. Wireless PCs successfully connect to these access points using WiFi password authentication and establish communication with PCs, laptops, printers, and remote campuses through the network infrastructure.

The successful results confirm correct access point configuration, proper wireless authentication, valid IP addressing, and fully functional wireless communication across the implemented university network.

12.5 Verification Results

After completing all testing procedures, the network is successfully verified, confirming that all campuses and departments communicate effectively without significant issues. The verification process ensures that the implemented design operates as intended across the entire system.

The results confirm correct IP configuration, proper subnetting implementation, successful WAN communication, and fully functional RIP routing between all routers. Both wired and wireless communication also operate correctly, validating the overall stability and reliability of the university network implemented in the Cisco Packet Tracer environment.

13 Wireless Communication

Wireless communication is implemented in selected campuses to provide flexible and secure network access without relying on physical cabling. PT access points are deployed in the Lahore and Quetta campuses to enable wireless devices to connect seamlessly with the university network infrastructure. This setup extends network accessibility beyond wired connections and supports efficient communication within departmental environments.

The wireless system enhances mobility by allowing devices to communicate with other departments and remote campuses using WiFi technology. Security is enforced through password protection, ensuring that only authorized devices can access the wireless network and maintaining controlled connectivity within the university system.

13.1 Access Point Connectivity

PT access points are installed in the Lahore and Quetta campuses to enable wireless communication within departmental networks, where each access point is connected to the departmental switch through an Ethernet link to integrate wireless and wired communication. This configuration allows wireless devices to interact seamlessly with the existing LAN infrastructure, ensuring smooth data exchange across the network.

Each access point is configured with a wireless network name (SSID), WiFi password, and appropriate IP addressing settings to ensure proper network integration and secure connectivity. Password protection is enabled to prevent unauthorized access and enhance wireless security. The access points successfully connect wireless devices such as PCs, laptops, printers, and other campus networks, enabling reliable and secure wireless communication across the system.

13.2 WIFI Connected PCs

Wireless PCs are connected to PT access points in the Lahore and Quetta campuses using WiFi communication to enable wireless access within the university network. Each wireless device is configured by selecting the appropriate wireless network, entering the WiFi password for authentication, and assigning a static IP address according to the departmental subnet design. This configuration ensures proper integration of wireless devices into the existing network infrastructure.

After successful authentication, wireless PCs establish stable connectivity and communicate effectively with wired PCs, laptops, printers, CCTV devices, and other campuses. This implementation enhances user mobility by allowing communication without physical cabling while maintaining reliable and secure network access across the system.

13.3 Wireless Data Sharing

Wireless data sharing is tested successfully between wireless and wired devices using Cisco Packet Tracer simulation mode to validate end-to-end connectivity across the network. The “Add Simple Protocol Data Unit” tool is used to generate test packets and verify that wireless communication functions correctly within the designed infrastructure.

Wireless devices successfully exchange data with devices within the same department, across different departments, and even between different campuses. The successful packet delivery confirms proper wireless connectivity, correct IP configuration, successful access point operation, and effective RIP-based routing between campuses. This verifies that wireless communication is fully functional and integrated as a reliable component of the overall university network design.

14 Applications of the System

The university network project has different practical applications in educational institutions and organizational environments. The network helps improve communication, resource sharing, security, and data transfer between campuses and departments. This system was designed to simulate a real-world university networking environment where multiple campuses communicate efficiently through wired and wireless technologies. The following are the major applications of the system:

14.1 Inter-Campus Communication

The university network enables communication between multiple campuses, including Karachi, Lahore, Quetta, and Peshawar, allowing students, staff, and departments to exchange information efficiently through WAN connectivity and RIP-based routing.

14.2 Departmental Communication

Each campus supports communication between departments such as Administration, Library, Examination, Accounts, and Education. Switches and routers ensure smooth data exchange and efficient internal networking within each campus environment.

14.3 Resource Sharing

The system supports sharing of resources across the network, including printers, files, data, and other network services. This reduces hardware dependency, improves operational efficiency, and enhances communication between departments.

14.4 Wireless Communication

PT access points provide wireless connectivity that allows users to access the network without physical cabling. Wireless PCs can securely connect using WiFi passwords, access resources, and communicate with other devices, improving flexibility and mobility within departments.

14.5 Security and Monitoring

CCTV cameras are deployed in all departments to enhance monitoring and security. They help in observing activities, improving campus safety, and maintaining a secure network environment across all locations.

14.6 Network Learning and Simulation

The project serves as a practical learning platform for understanding key networking concepts such as IP addressing, subnetting, routing, wireless networking, and network topology design using Cisco Packet Tracer, bridging theoretical knowledge with real-world application.

14.7 Data Communication and Testing

The network supports reliable data transmission between wired devices, wireless devices, departments, and campuses. Connectivity testing using the Simple Protocol Data Unit tool confirms successful packet delivery and validates proper network operation across the entire system.

15 Strengths and Weaknesses

Every network system has some advantages and limitations. This university network project also contains several strong points that improve communication and network management, along with some weaknesses that can be improved in future development.

The strengths and weaknesses of the project are discussed below.

15.1 Strong Points

- **Multi-Campus Connectivity:** All university campuses are interconnected through WAN serial links and communicate successfully using RIP dynamic routing. This improves inter-campus communication, data sharing, and overall network connectivity across the system.
- **Proper Subnetting Design:** A structured subnetting approach is implemented by assigning separate /28 subnets to each department. This reduces unnecessary traffic, improves network organization, and ensures efficient IP address management.
- **Dynamic Routing using Routing Information Protocol (RIP):** The RIP routing protocol automatically exchanges routing information between routers, reducing manual configuration. It enables automatic route learning, improves communication between campuses, and simplifies overall network management.
- **Wired and Wireless Communication:** The network supports both wired and wireless communication. PT access points provide wireless connectivity that enhances user mobility, flexibility, and resource sharing across the university environment.
- **Security through WiFi Passwords:** Wireless networks are secured using WiFi passwords to ensure controlled access. This improves network security, user authentication, and protection against unauthorized devices.
- **Resource Sharing:** The system enables efficient sharing of printers, data, and other network resources across departments and campuses. This improves operational efficiency and reduces hardware costs within the university network.

15.2 Weak Points

Although the project works successfully, there are some limitations and weaknesses that can be improved in the future.

- **Limited Network Security:** The current implementation uses only basic security mechanisms, while advanced security features such as firewalls, access control lists (ACLs), and encryption techniques are not included. This limits the overall protection level of the network against unauthorized access and potential threats.
- **Use of Routing Information Protocol (RIP):** RIP is implemented due to its simplicity and ease of configuration; however, it is not suitable for large-scale networks because it has a limited hop count, slower convergence, and reduced scalability. More advanced protocols such as OSPF or EIGRP would provide better performance and efficiency in complex environments.
- **Static IP Addressing:** The network relies on manually configured static IP addresses for all devices, which increases configuration time and the risk of IP conflicts. In larger networks, this approach reduces efficiency, and the use of DHCP would improve automation and simplify network management.
- **Limited Wireless Implementation:** Wireless communication is implemented only in selected campuses and departments, meaning that full wireless coverage is not provided across the entire network. A more extensive wireless deployment would improve flexibility and accessibility.
- **Simulation-Based Environment:** The project is developed in Cisco Packet Tracer, which provides a simulated environment rather than a real-world deployment. A practical implementation would require physical networking hardware, advanced security systems, higher bandwidth capacity, and professional monitoring tools for real-time management.
- **Limited Scalability:** The current design is suitable for small to medium-sized university networks. If the number of campuses, departments, or users increases significantly, the network would require redesign along with more advanced routing, addressing, and security mechanisms to maintain performance and stability.

16 Future Enhancements

16.1 Implementation of Advanced Routing Protocols

The current network uses RIP for inter-campus communication; however, future improvements can include advanced routing protocols such as OSPF (Open Shortest Path First) and EIGRP (Enhanced Interior Gateway Routing Protocol). These protocols provide faster convergence, improved scalability, better performance, and more efficient path selection in larger and more complex network environments.

16.2 Dynamic Host Configuration Protocol (DHCP) Implementation

Dynamic Host Configuration Protocol (DHCP) is implemented within the university network to support automatic IP address assignment for connected devices. DHCP allows devices to obtain valid network configuration dynamically without requiring manual IP assignment, improving efficiency and reducing administrative workload.

The network design incorporates both static and dynamic IP addressing methods to maintain a balanced and flexible communication environment. Static addressing is applied to critical network infrastructure devices for reliable connectivity, while DHCP-based dynamic addressing is used for selected client devices to simplify network administration, reduce configuration errors, and improve scalability across the university network.

16.3 Improved Network Security

Additional security mechanisms can be integrated to strengthen network protection. These may include firewalls, access control lists (ACLs), secure wireless encryption, and user authentication systems, which collectively enhance defense against unauthorized access and network-based threats.

16.4 Internet Connectivity

Currently, the network is limited to internal communication between campuses and departments. Future improvements may include internet access, cloud services, and web-based applications to support external communication and expand functionality for university users.

16.5 Expansion of Wireless Communication

Wireless connectivity is currently available only in selected campuses. In future development, full campus-wide WiFi coverage, additional access points, and more wireless devices can be deployed across all campuses to enhance mobility and communication flexibility.

16.6 Network Monitoring System

Advanced monitoring tools can be implemented to observe network performance, detect issues, analysed traffic, and improve troubleshooting. This would enhance reliability, optimize bandwidth usage, and improve overall network stability.

16.7 Real Hardware Implementation

The current project is developed in a simulation environment using Cisco Packet Tracer. Future work can involve real physical networking devices such as routers, switches, and wireless equipment to provide practical deployment experience and improve system reliability in real-world conditions.

17 Conclusion

This project successfully designs and implements a multi-campus university network using Cisco Packet Tracer. The network interconnects four campuses, including Karachi, Lahore, Quetta, and Peshawar, through WAN serial links and RIP dynamic routing, enabling seamless communication across geographically distributed locations.

Separate subnetworks are created for each department using subnetting techniques to ensure proper segmentation and efficient network management. Various networking devices, including routers, switches, access points, PCs, laptops, printers, and CCTV cameras, are successfully configured within the network. Static IP addressing is applied to all devices, while RIP routing enables automatic route exchange between campuses. Wireless communication is

implemented in the Lahore and Quetta campuses using PT access points with password protection to ensure secure connectivity.

Connectivity testing using the Simple Protocol Data Unit tool confirms successful communication between departments, campuses, wired devices, and wireless devices. The project provides practical understanding of key networking concepts such as IP addressing, subnetting, routing, wireless networking, network topology design, and Cisco Packet Tracer simulation.

Overall, the project achieves its objectives by delivering an organized, efficient, and realistic university networking environment that effectively demonstrates core principles of modern computer networking.

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